OCEANEYE · 2026 世界职业院校技能大赛 · WORLD VOCATIONAL COLLEGE SKILLS COMPETITION

OceanEye

AI视觉驱动的水下机器人智能生物检测系统

AI Vision-driven Underwater Robot Intelligent Biological Detection System

国际赛道 International Track | 东南亚赛区 Southeast Asia Region | 柬埔寨政策背景 Cambodia Policy

绿色 Green · 数字 Digital · 渔业 Fisheries · 智能AI Artificial Intelligence

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柬埔寨国家战略：绿色发展与数字化转型双重驱动

Cambodia National Strategy: Green Development & Digital Transformation

五角战略第一阶段

Pentagonal Strategy - Phase I

围绕“增长、就业、公平、效率、可持续”五大发展目标，强调可持续、包容和韧性的国家发展路径，为绿色技术与数字应用提供政策土壤。

Centred on five goals — growth, employment, equity, efficiency and sustainability — charting a sustainable, inclusive and resilient path that nurtures green technology and digital adoption.

数字经济与社会政策框架

Digital Economy and Society Policy Framework 2021-2035

提出建设充满活力的数字经济与社会，推动国家、企业和社会各主体的数字采用与数字转型，为 AI 视觉系统落地提供数字基础设施支撑。

Aims to build a vibrant digital economy and society, driving digital adoption and transformation across government, business and citizens — the digital foundation for AI vision deployment.

绿色可持续	数字政府	数字企业	数据治理	技术应用
Sustainability	Digital Government	Digital Business	Data-driven Governance	AI & Digital Adoption

资料来源 Source: Royal Government of Cambodia, Pentagonal Strategy Phase I; Cambodia Digital Economy and Society Policy Framework 2021-2035

柬埔寨生态与渔业政策：从环境治理到智能监测

Environment & Fisheries Policy: From Governance to Intelligent Monitoring

环境循环战略 2023—2028

Circular Strategy on Environment 2023-2028

CLEAN
清洁
污染控制与测量体系现代化
Modernising pollution control & measurement

GREEN
绿色
自然资源保护与生态环境治理
Nature conservation & eco-governance

SUSTAINABLE
可持续
提升环境治理能力与可持续发展水平
Strengthening governance capacity & sustainability

对 OceanEye 的要求：低成本、连续化、数据化、可视化的水域环境监测能力。

What it demands of OceanEye: low-cost, continuous, data-driven and visual water-environment monitoring.

柬埔寨渔业治理

Cambodia Fisheries Governance

516

个社区渔业组织 Community Fisheries

淡水 472 个 / 海洋 44 个 · 472 Freshwater / 44 Marine

传统方式 Traditional Approach
人工巡查、看视频、人工统计
Manual patrol, video review & counting
效率低 · 连续性不足 Low efficiency & poor continuity

智能方式 Smart Approach
水下摄像 + AI复原 + 自动检测
Underwater camera + AI restoration + auto detection
自动识别 · 连续监测 Auto recognition & continuous monitoring

资料来源 Source: Cambodia Ministry of Environment, Circular Strategy on Environment 2023-2028; Cambodia Fisheries Administration / MAFF

从政策目标到场景需求：OceanEye 的项目切入

From Policy Goals to Scenario Needs: OceanEye Entry Point

政策目标 / Policy Goal	应用场景 / Scenario	OceanEye 能力 / Capability
可持续渔业 Sustainable Fisheries	水产养殖监测 Aquaculture Monitoring	生物目标识别 / 数量统计 Biological ID & Counting
生态保护 Ecological Protection	海洋生态监测 Marine Ecology Monitoring	低质图像复原 / 视觉数据增强 Low-quality Restoration / Enhancement
数字化治理 Digital Governance	社区渔业管理 Community Fishery Management	连续化数据采集 / 可视化输出 Continuous Data Collection / Visualization
安全巡检 Safety Inspection	水下设施检测 Underwater Facility Inspection	异常目标定位 / 辅助预警 Anomaly Localization / Early Warning

技术痛点：水下图像“看不清”、AI“识不准”

Technical Pain Points: Low Visibility & Unreliable Detection

痛点一：水下图像退化

Pain Point I: Image Degradation

光照吸收 Light Absorption

红光快速衰减，蓝绿色偏色

Rapid red-channel attenuation causes blue-green casts

水体散射 Water Scattering

边缘模糊，局部对比度下降

Blurred edges and reduced local contrast

纹理缺失 Texture Loss

细节弱化，可判别特征不足

Weakened details, insufficient discriminative features

痛点二：目标检测困难

Pain Point II: Detection Difficulties

小目标多 Numerous Small Objects

海参、海胆、扇贝、海星等目标尺寸小、轮廓弱

Sea cucumbers, urchins, scallops & starfish: small size, weak contours

背景复杂 Complex Backgrounds

水草、泥沙、礁石与目标颜色相近，干扰强

Seaweed, sediment & reefs share similar colours, strong interference

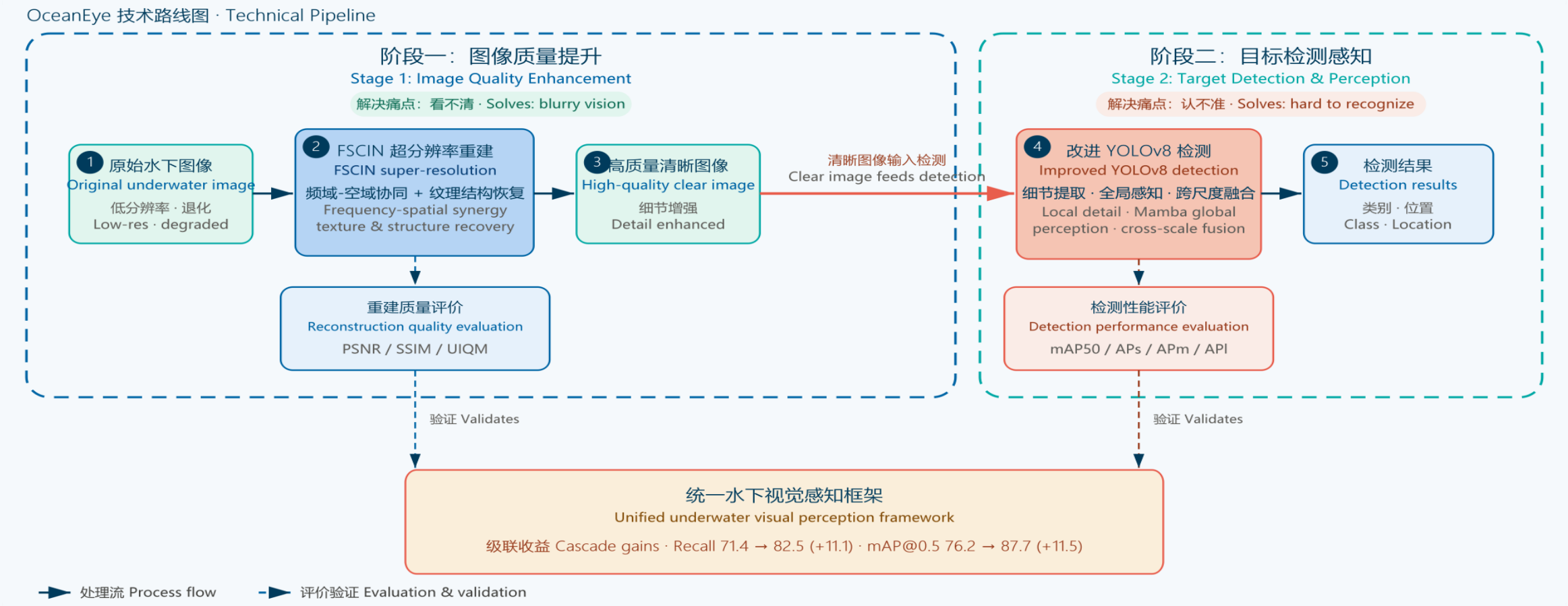
边界模糊 Blurred Boundaries

低对比度输入导致漏检、误检、定位偏移

Low-contrast input causes misses, false alarms & localization offsets

总体技术路线：复原增强与目标检测级联

Overall Technical Roadmap: Restoration-Detection Cascade



低质量水下图像 Low-quality Input → **FSCIN 图像复原** Image Restoration → **增强图像** Enhanced Image → **改进 YOLOv8 检测** Improved YOLOv8 Detection → **检测输出** Detection Output

PART 01

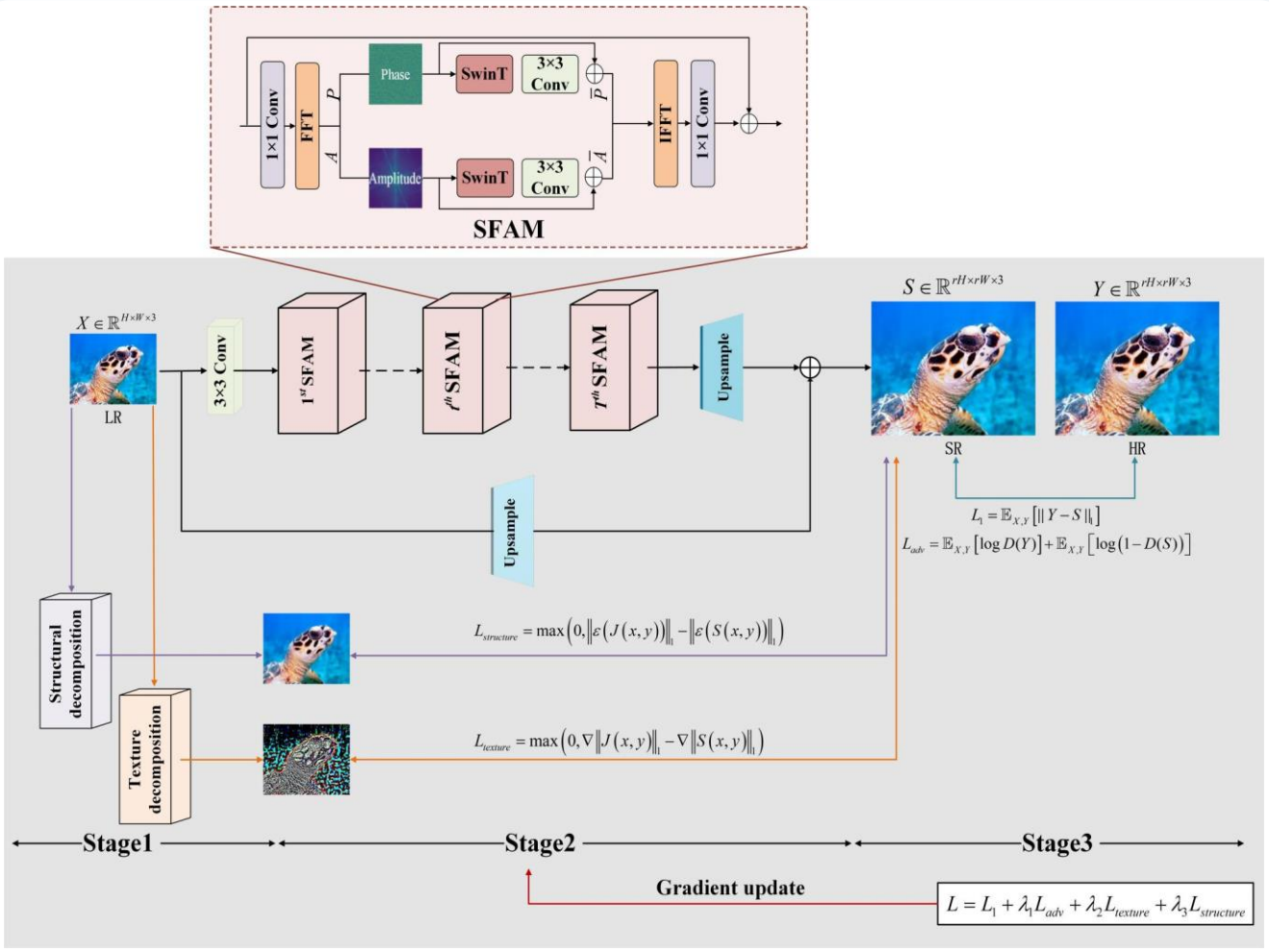
水下目标增强技术

Underwater Target Enhancement Technology

主讲人 Presenter A

FSCIN 复原模型：频率域与空间域协同建模

FSCIN Model: Frequency-Spatial Collaborative Modeling



空间特征提取

Spatial Feature Extraction

捕获局部纹理与边缘信息 · Captures local texture & edges

频域特征建模

Frequency Feature Modeling

FFT/IFFT 分析幅度谱与相位谱 · Amplitude & phase spectrum analysis

SFAM 融合与重建

Spatial-Frequency Attention Module

自适应融合频域先验与空间纹理 · Adaptive fusion of frequency prior & spatial texture

纹理-结构联合损失

Texture-Structure Joint Loss

兼顾细节恢复与结构保真 · Balances detail recovery & structural fidelity

图像复原实验：性能与效率兼顾

Restoration Experiments: Performance & Efficiency

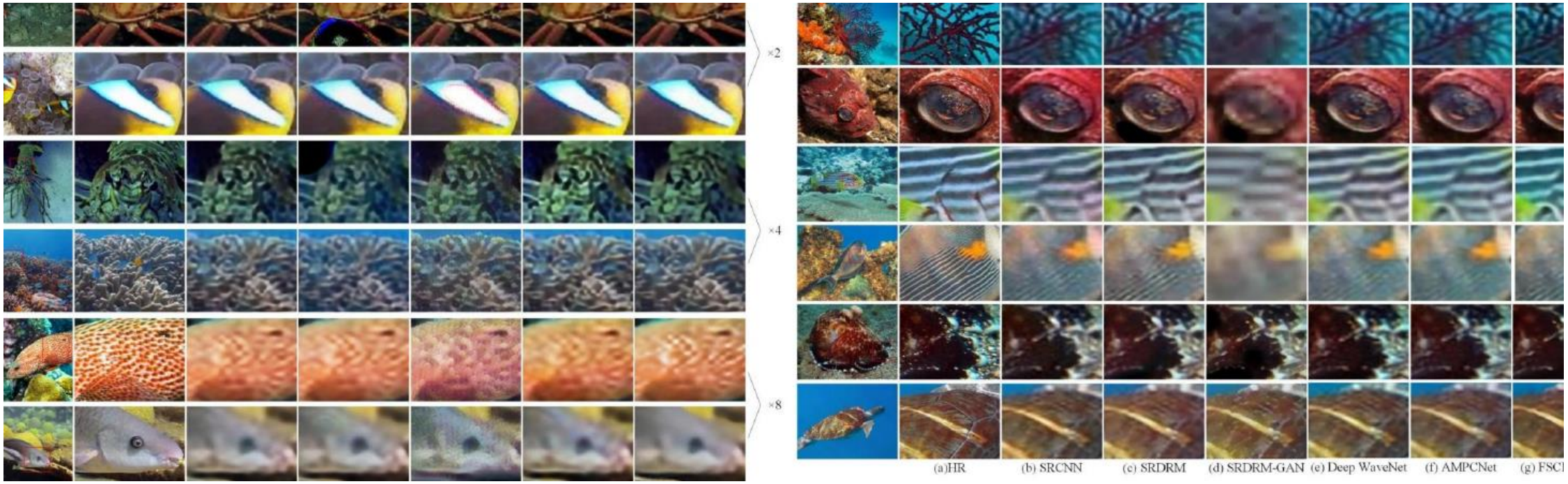
30.04
USR-248 ×2 超分 PSNR
SR PSNR on USR-248 ×2

25.98
UFO-120 ×2 超分 PSNR
SR PSNR on UFO-120 ×2

24.03
×8 超分 PSNR
SR PSNR at ×8 scale

0.55M
参数量
Parameters

30.70G
计算量
FLOPs



PART 02

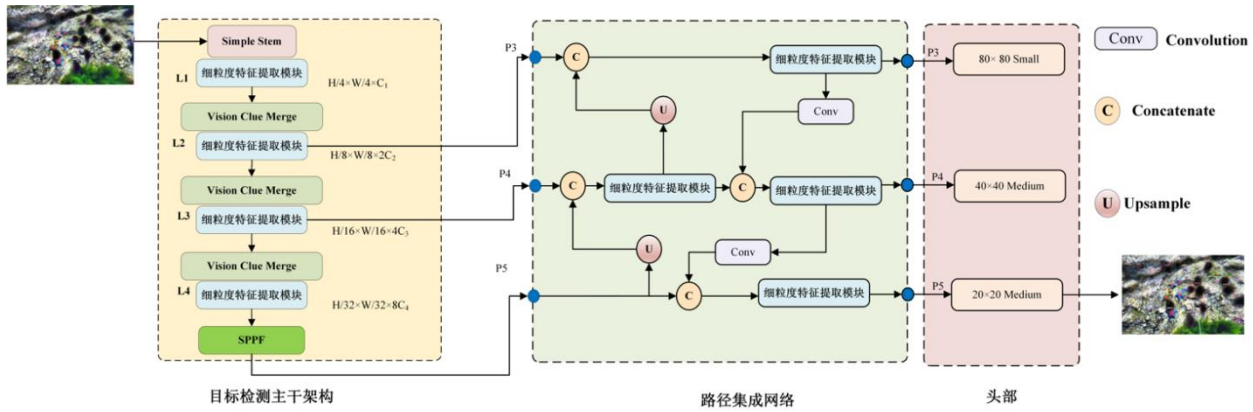
水下目标检测技术

Underwater Target Detection Technology

主讲人 Presenter B

改进 YOLOv8：局部细节、全局上下文与多尺度融合

Improved YOLOv8: Local Detail, Global Context & Multi-scale Fusion



CAEM

Channel-Spatial Attention Module

通道-空间注意力，聚焦边缘纹理

Channel-spatial attention focusing on edge textures

VSSM

Vision State Space Model

Mamba 全局感知，建模长程依赖

Mamba global perception for long-range dependencies

MCFM

Multi-scale Cross Fusion Module

多尺度交叉融合，提升小目标召回

Multi-scale cross fusion boosting small-object recall

目标检测实验：综合精度与小目标能力提升

Detection Experiments: Overall Precision & Small-object Ability

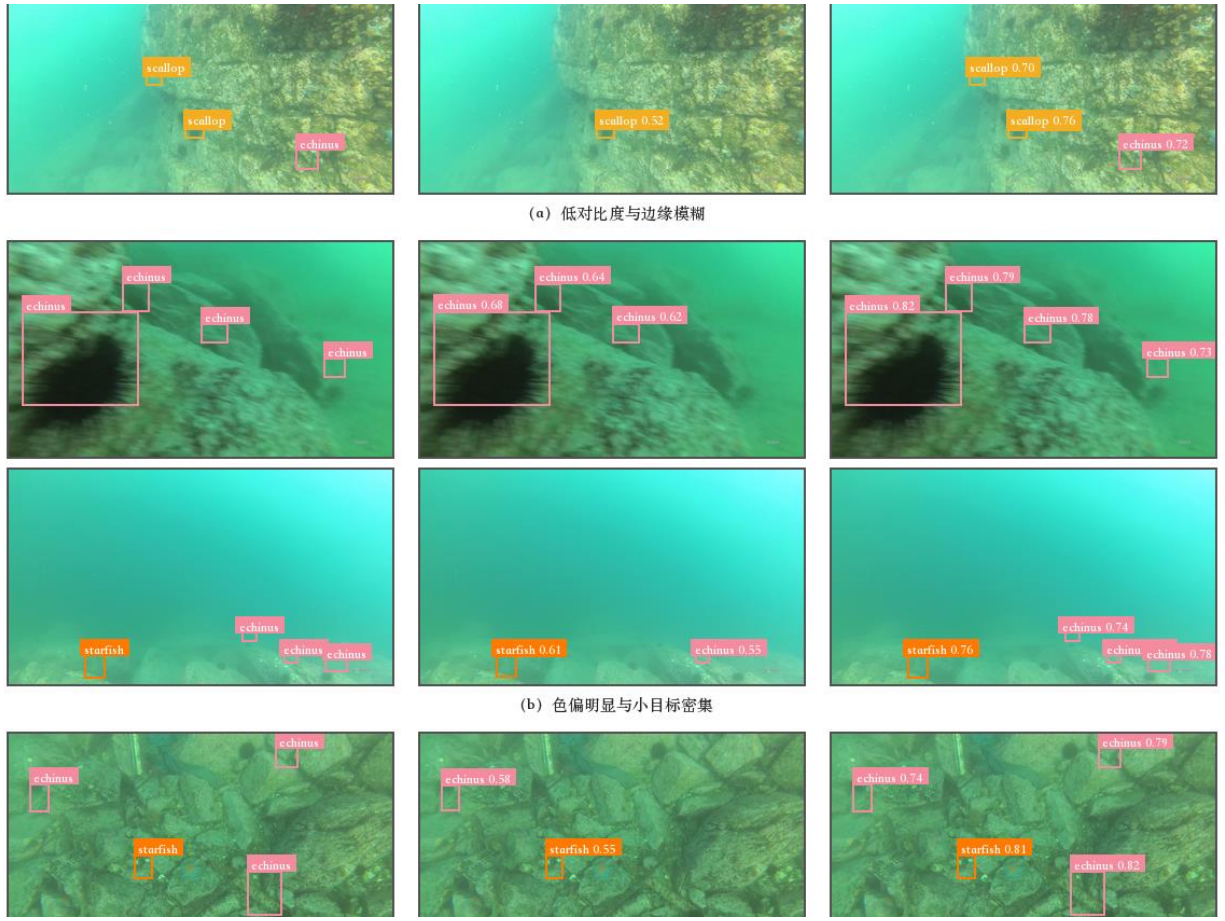
主指标对比 / Main Metrics

YOLOv8 基线 Baseline mAP50	改进 YOLOv8 Improved mAP50
85.5	87.3

YOLOv8 基线 Baseline APs	改进 YOLOv8 Improved APs
16.6	17.7

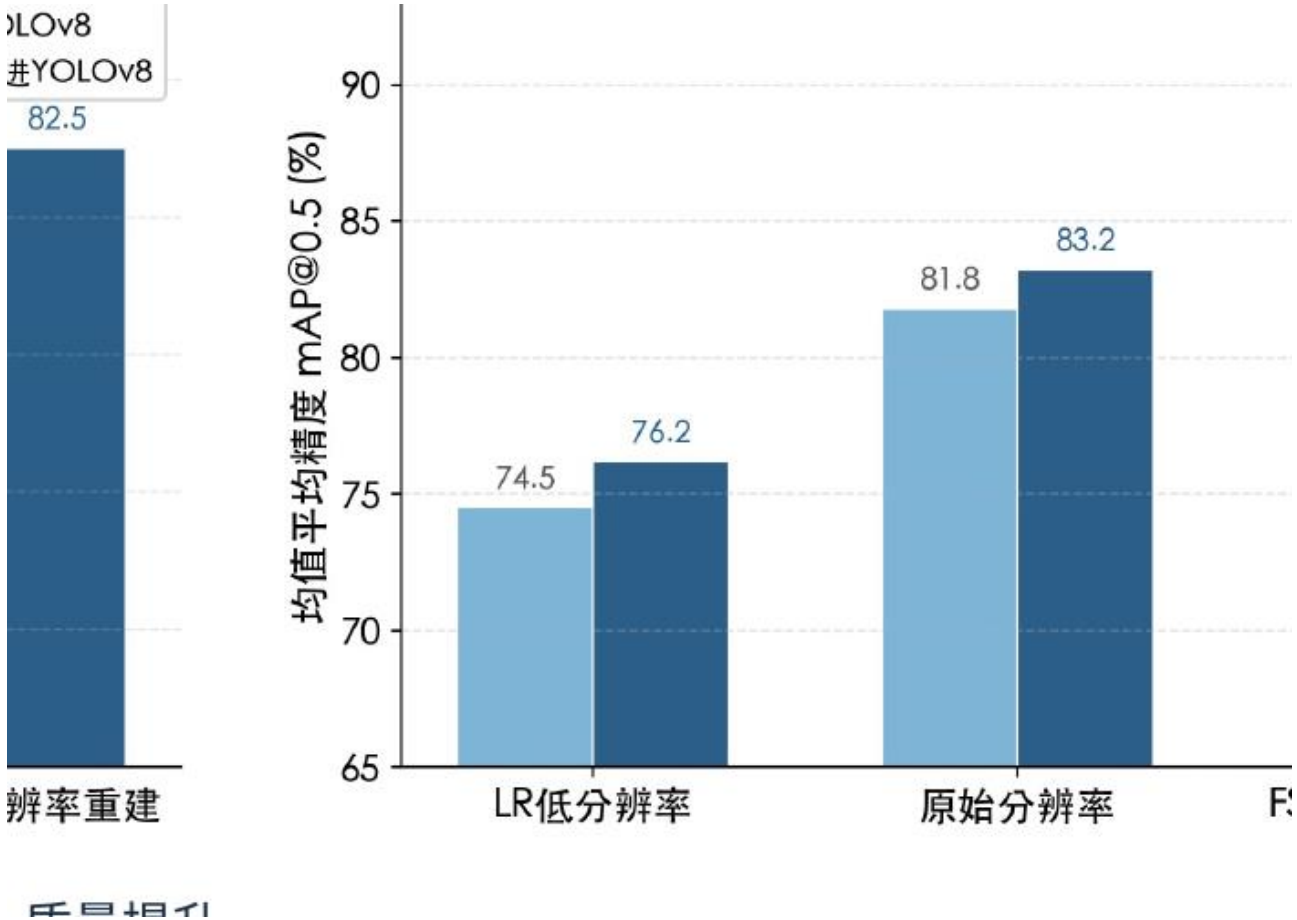
消融结论 / Ablation Insights

+ CAEM：边缘纹理增强 · Edge-texture enhancement
+ VSSM：全局依赖建模 · Global dependency modeling
+ MCFM：跨尺度特征融合 · Cross-scale feature fusion



复原—检测级联收益：从图像质量到任务性能

Restoration-Detection Cascade Gains: From Quality to Task Performance



LR 基线 Recall 召回率

71.4 → 82.5

提升 +11.1 pts · Improvement

LR 基线 mAP@0.5 平均精度

76.2 → 87.7

提升 +11.5 pts · Improvement

核心结论 Key Takeaway

图像复原不是“美化”，而是为下游检测恢复可判别特征。

Restoration is not “beautification” — it recovers discriminative features for downstream detection.

系统应用：面向柬埔寨水域的四大场景

System Application: Four Scenarios for Cambodian Waters

01

水产养殖监测

Aquaculture Monitoring

面向洞里萨湖等养殖区，实现生物目标识别与数量统计。

Bio-target recognition & counting for aquaculture zones like Tonlé Sap.

02

海洋生态监测

Marine Ecology Monitoring

低质图像复原与视觉增强，支撑珊瑚、鱼类资源调查。

Restoration & enhancement supporting coral and fish-stock surveys.

03

社区渔业管理

Community Fishery Management

连续化数据采集与可视化输出，服务 516 个社区渔业组织。

Continuous data collection & visualization serving 516 community fisheries.

04

水下设施巡检

Underwater Facility Inspection

异常目标定位与辅助预警，保障养殖设施与水利工程安全。

Anomaly localization & early warning for aquaculture and waterworks safety.

创新点与总结

Innovations & Summary

01

复原增强

Restoration Enhancement

构建频率-空间协同的水下图像复原模型 FSCIN，在纹理细节恢复与结构保真之间取得平衡。

FSCIN balances texture-detail recovery and structural fidelity in frequency-spatial collaborative restoration.

02

检测优化

Detection Optimization

在 YOLOv8 中引入 CAEM、VSSM、MCFM 三模块，针对水下小目标与模糊边界提升检测精度。

CAEM, VSSM & MCFM integrated into YOLOv8 to boost accuracy for small objects and blurred boundaries.

03

任务闭环

Task-level Closed Loop

将图像复原结果直接转化为检测增益，形成面向柬埔寨水域应用的端到端智能感知方案。

Restoration results translate directly into detection gains — an end-to-end perception solution for Cambodian waters.

核心结论 Key Takeaway

OceanEye 让水下图像从“可见”走向“可用”，让目标检测从“能识别”走向“更稳定”，契合柬埔寨绿色、数字、渔业发展战略。

OceanEye turns underwater images from “visible” to “usable”, and detection from “recognizable” to “stable” — aligned with Cambodia’s green, digital and fishery strategies.

相关成果展示

Achievements & Recognitions



竞赛获奖 / Competition Awards



软件著作权 / Software Copyright



专利受理 / Patent Application



成果支撑 / Supporting Achievements

感谢聆听

Thank You for Your Attention

恳请批评指正 · Comments and suggestions are welcome